

Instructions



How is it going ? Do you have problems with practical tasks ? Any concerns with learning materials ?

problem wit sharing repo

have to upload only html

i think no

fine

all fine

haven

good but a bit tired

thats it

in interface design we

transform to htm js form

Today's agenda

- Total Recall
- ORM & JPA & LIFECYCLE
- JPA Example
- Quiz
- Practical Task Discussion

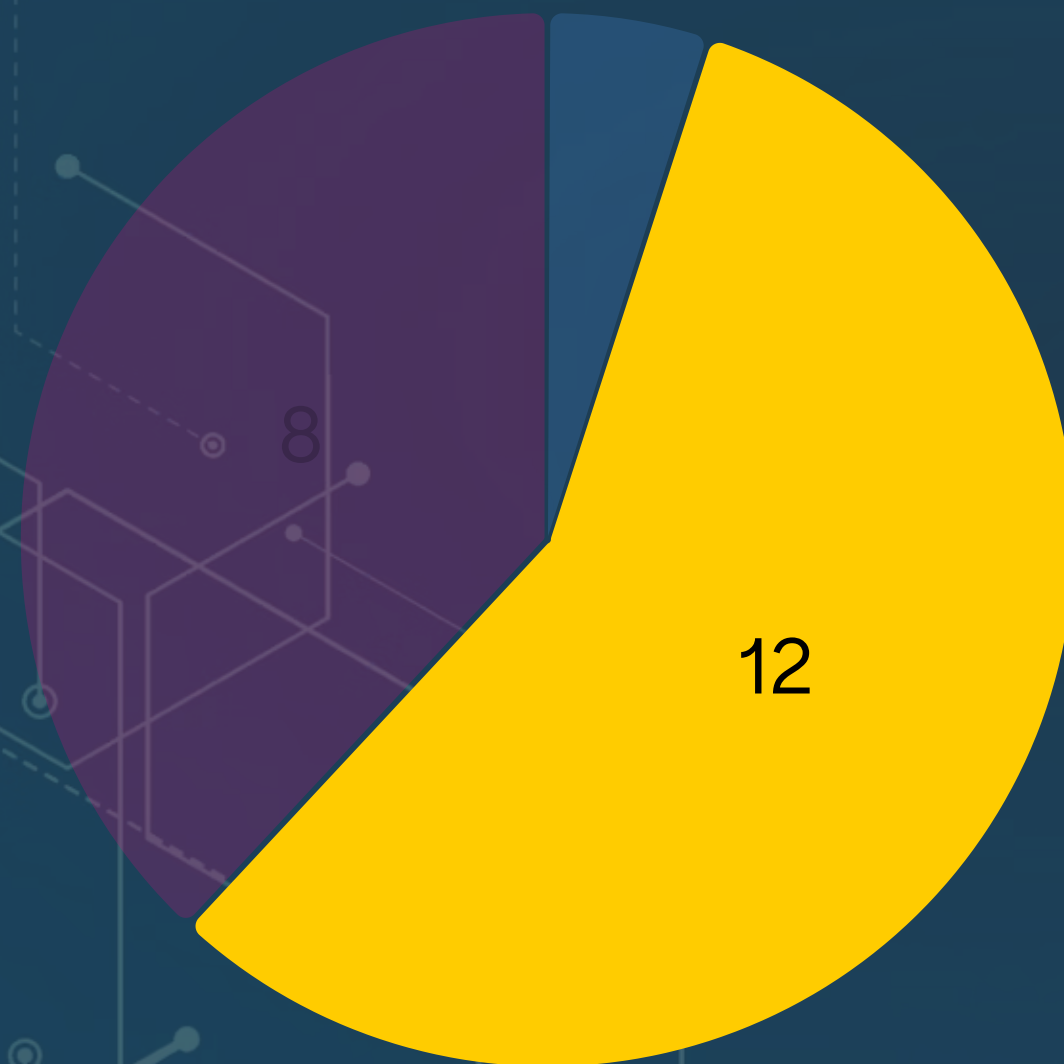
Total Recall



What do you remember from the last session?

loveable
ui builder io you
vibe coding figma mvc-vm-p

Who is receiving requests in MVVM?



1	Model	✗
12	View	✓
8	View-Model	✗

Select most correct meaning:

16 ✓



mandatory many

4 ✗



many to one

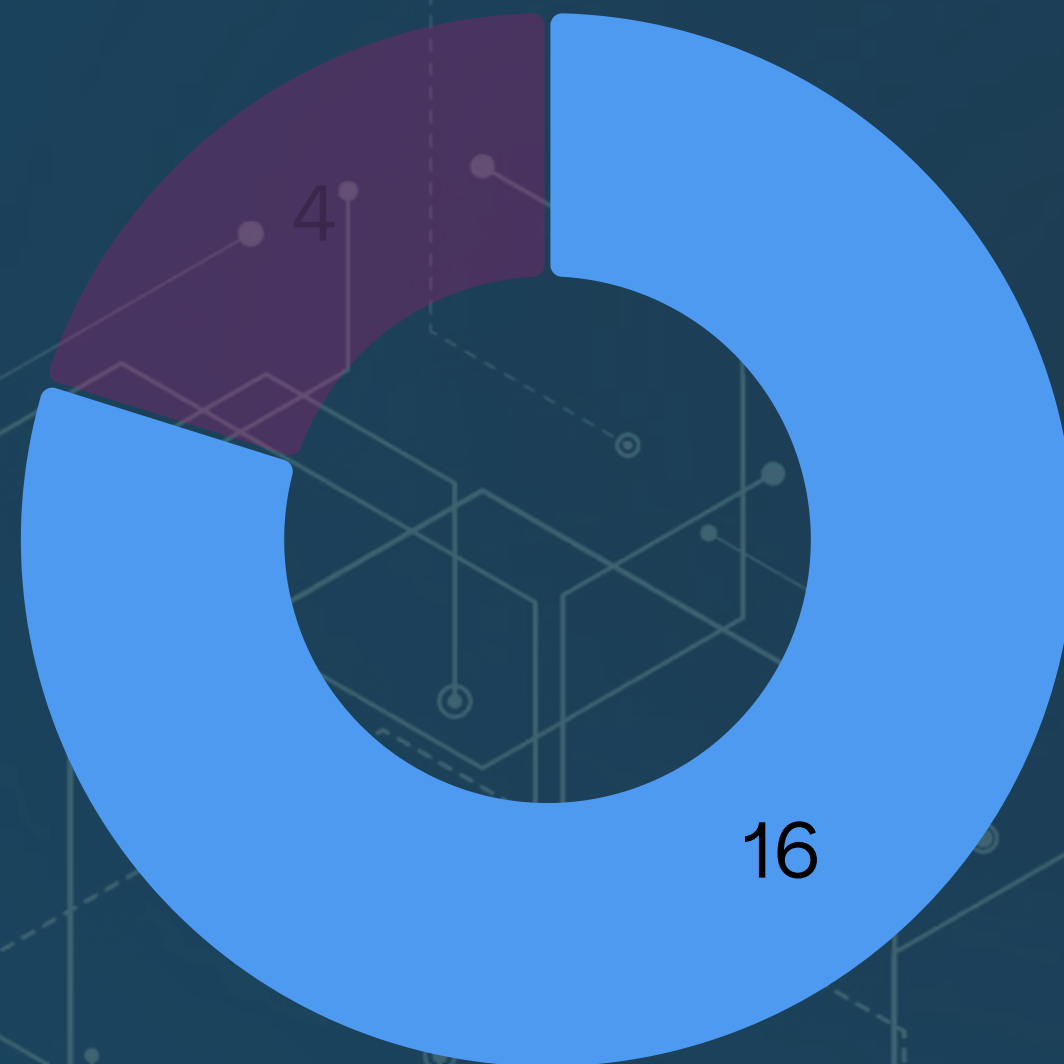
2 ✗



optional many



How can designers assist users in making decisions when encountering a new interface?



16

Add emphasis to the "next step" to guide users through their most likely task



0

Overwhelm users with multiple options to encourage exploration

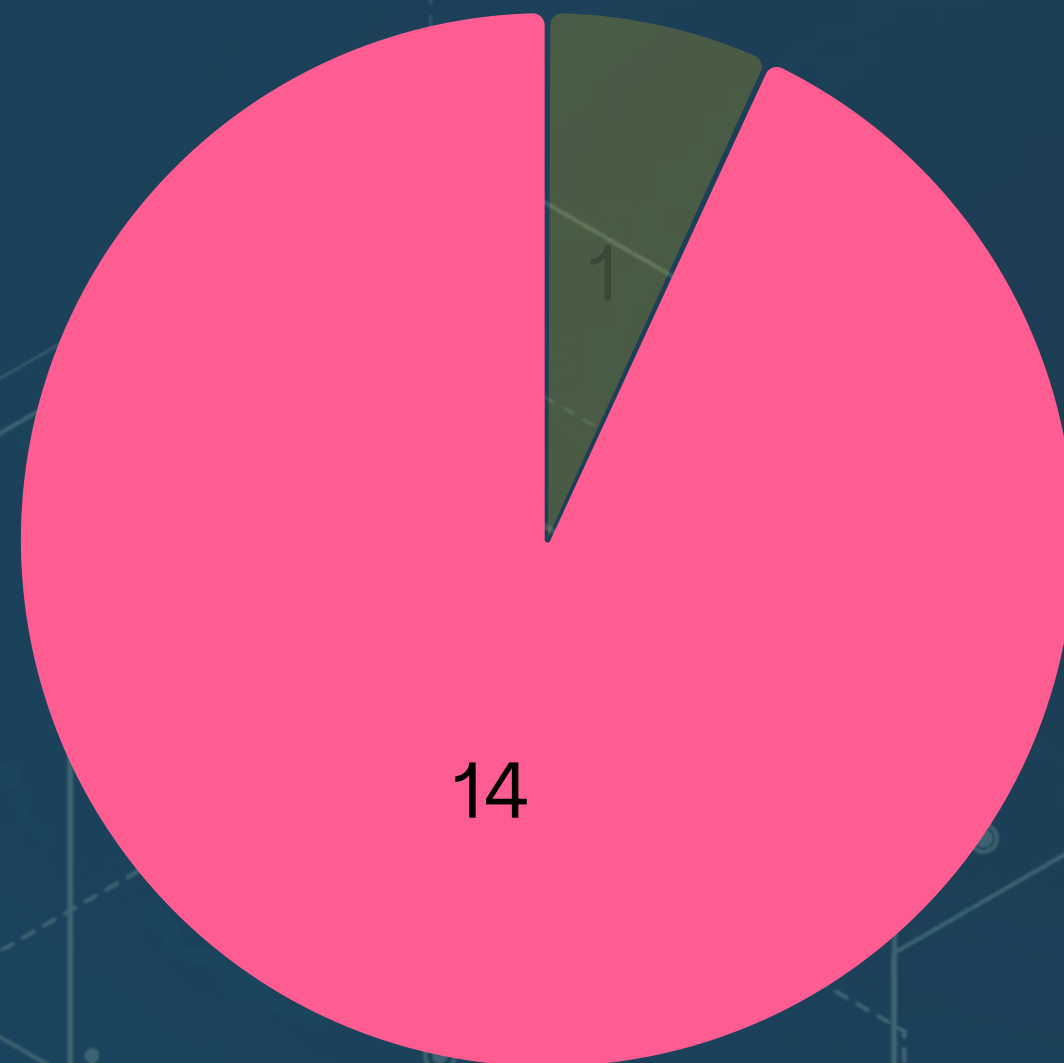


4

Include all features in the initial interface design without considering user progression



As of recommendations, how can UI designers **direct users' attention** effectively?

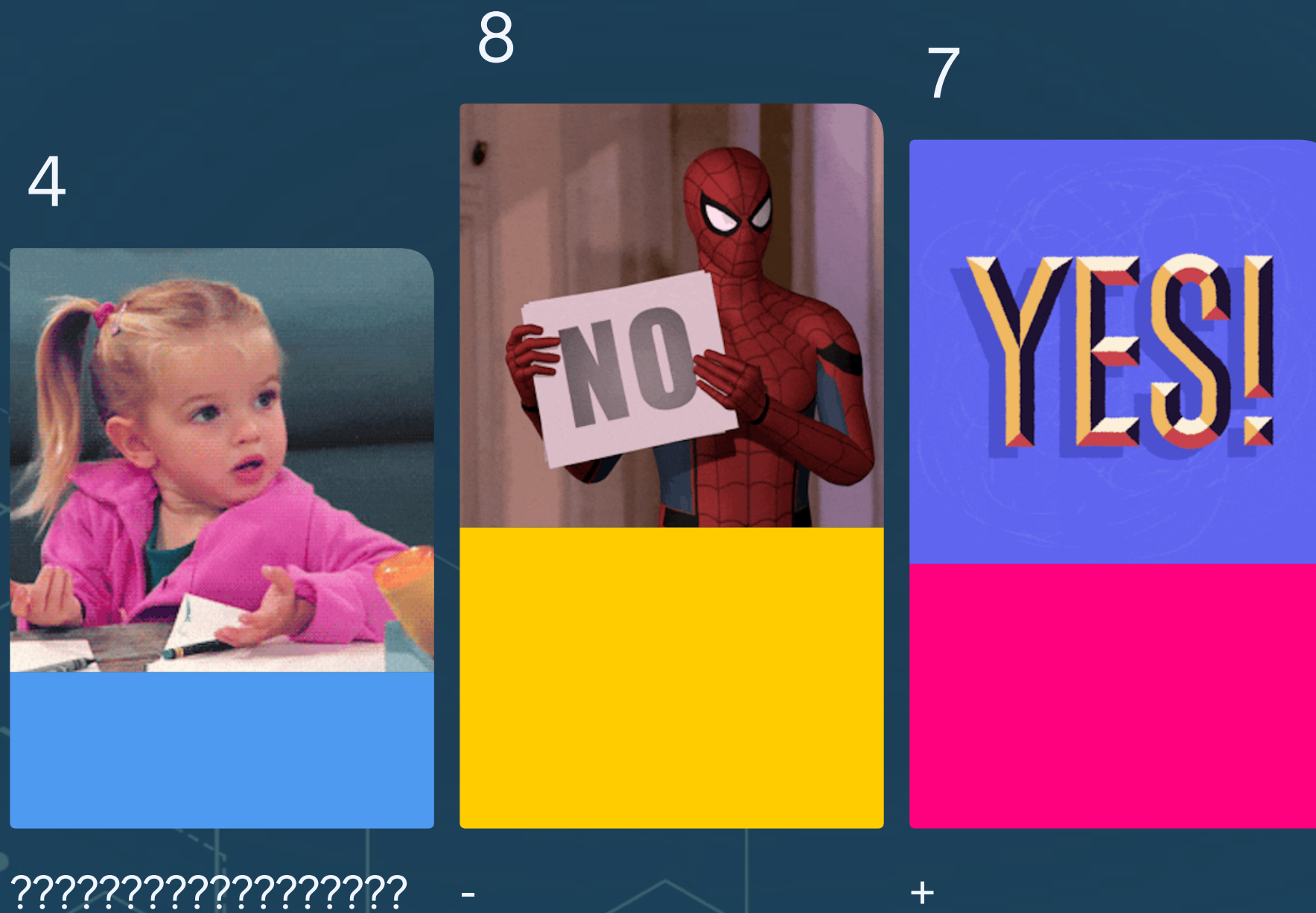


- 0 By incorporating random colors and fonts ✗
- 1 By making the UI as busy as possible ✗
- 0 By avoiding changes in size, placement, and color ✗
- 14 By using changes in size, placement, fonts, or color of elements in the page layout ✓

JPA (ws-05)

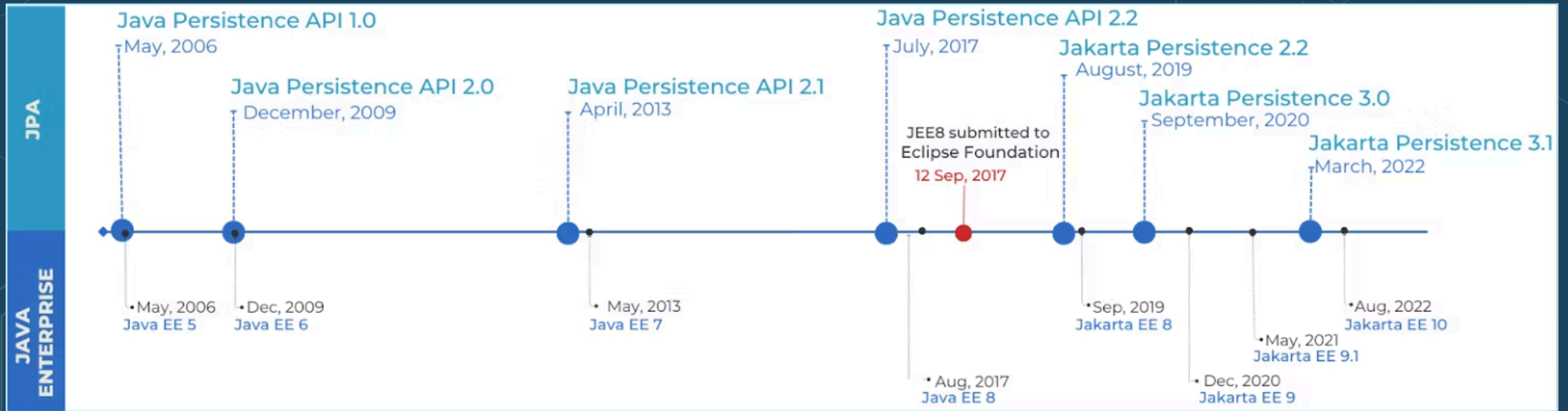
ORM | JPA | ENTITY | PERSISTENCE CONTEXT

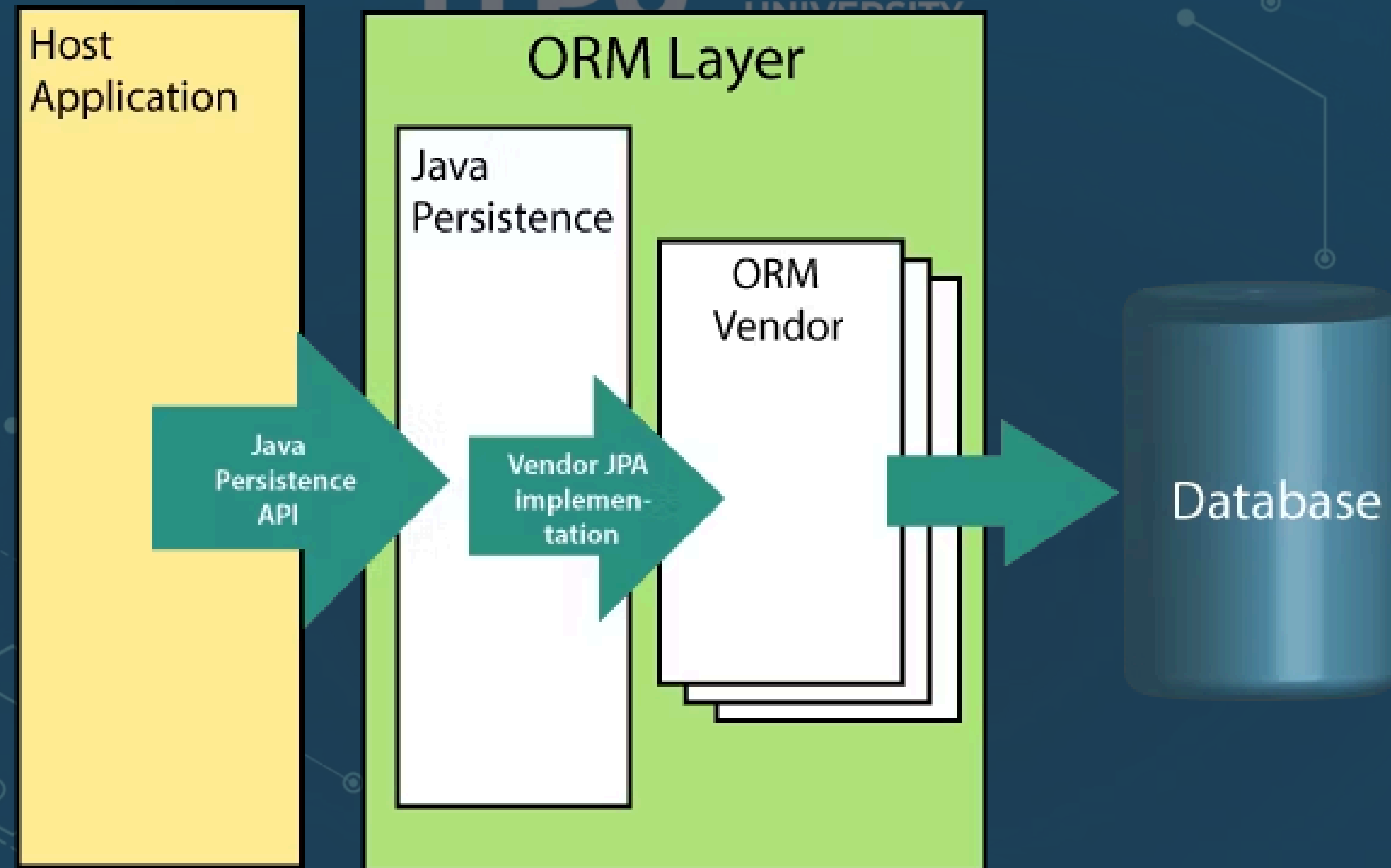
Have you started the theory of 'JPA'?



Definitions

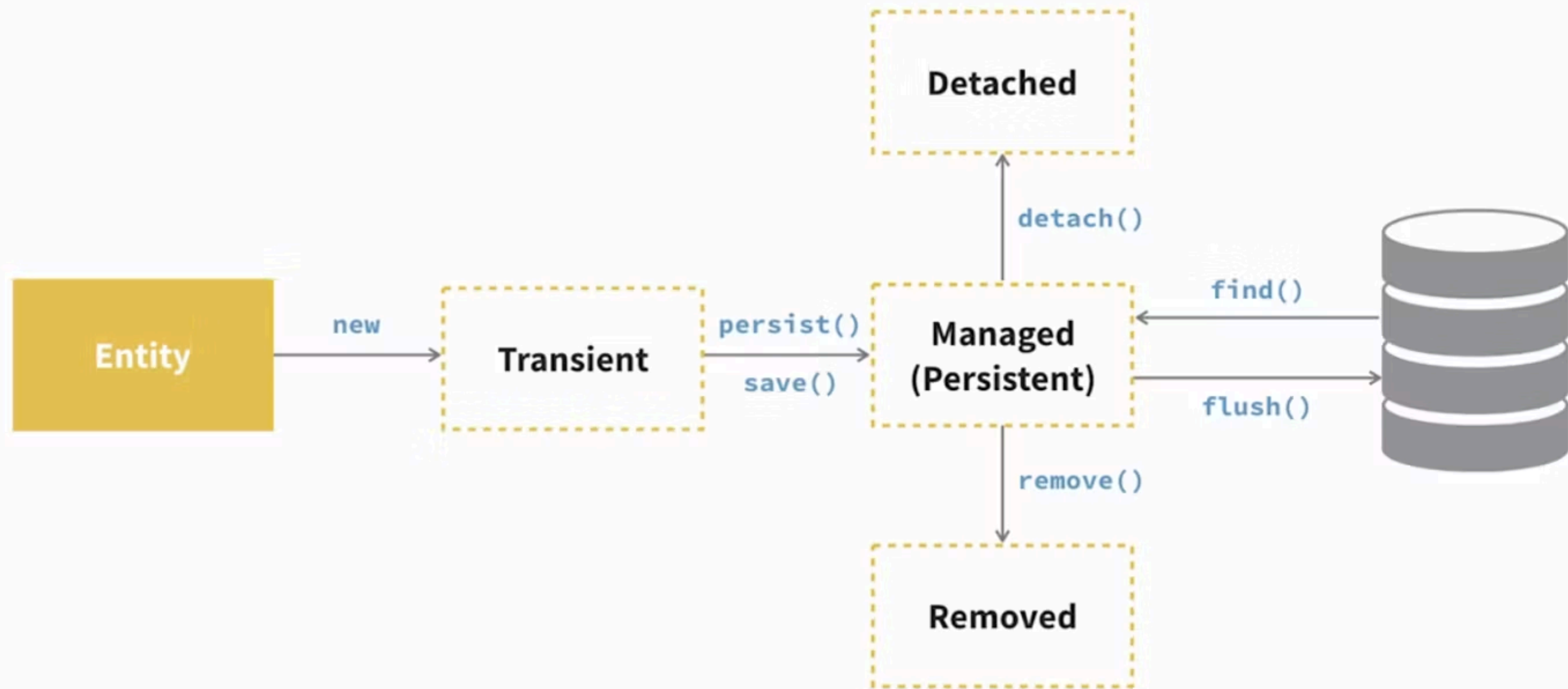
- ORM == Object Relational Mapping
- JPA == Java Persistence API || Jakarta Persistence API
- javax.* --->>> jakarta.*
- Vendor == Hibernate || etc.





process flow

Entity Lifecycle



Explore module ws-05

<https://github.com/AndreiRohau/itpu-web-development-with-java-2026>

10 mins beak

Quiz

Which annotation is used to declare an *Entity* class in JPA?

18 ✓

1 ✗

Database

0 ✗

Table

Entity

Which file is used to configure *persistence* units in a JPA application?

20 ✓



0 ✗

web.xml

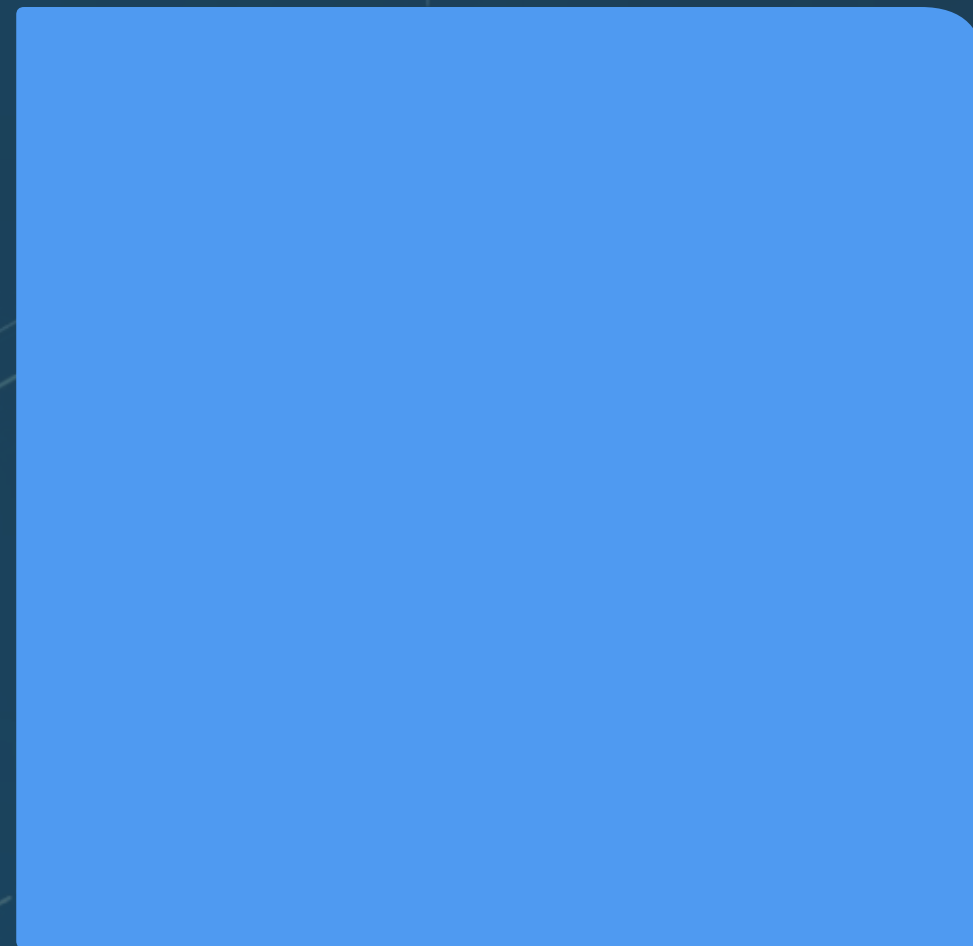
persistence.xml

1 ✗

applicationContext.xml

What is the purpose of **EntityManager** in JPA?

20 ✓



To manage the persistence context
and perform database operations

0 ✗

To handle user authentication

1 ✗

To manage network connections

Which JPA annotation is used to specify the *primary key* of an entity?

2 ✖

PrimaryKey

0 ✖

Primary

18 ✔

Id

What does the mappedBy attribute specify in JPA annotations?

21 ✓



0 ✗

The name of the database table

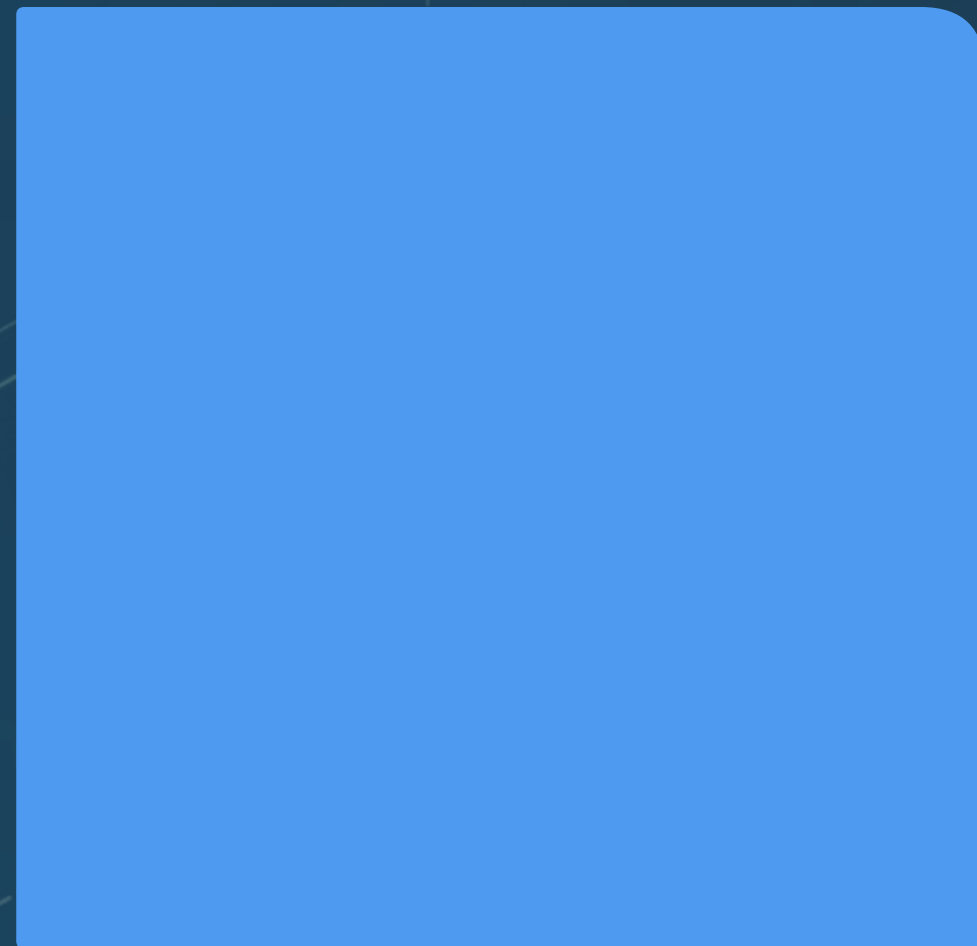
0 ✗

The foreign key column name

The owning side of a bidirectional relationship

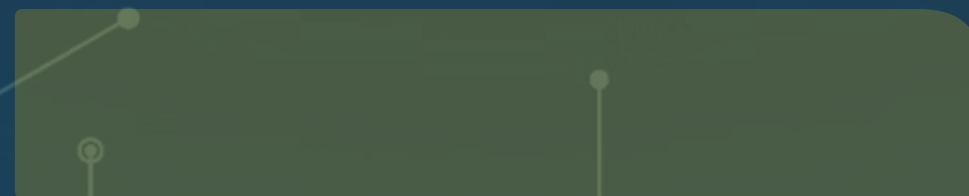
Imagine Employee has @OneToOne relation with Profile, and Employee is *owning* relation. Select correct statemet from below:

16 ✓



Employee table will have profile_id

3 ✗



Profile table will have employee_id

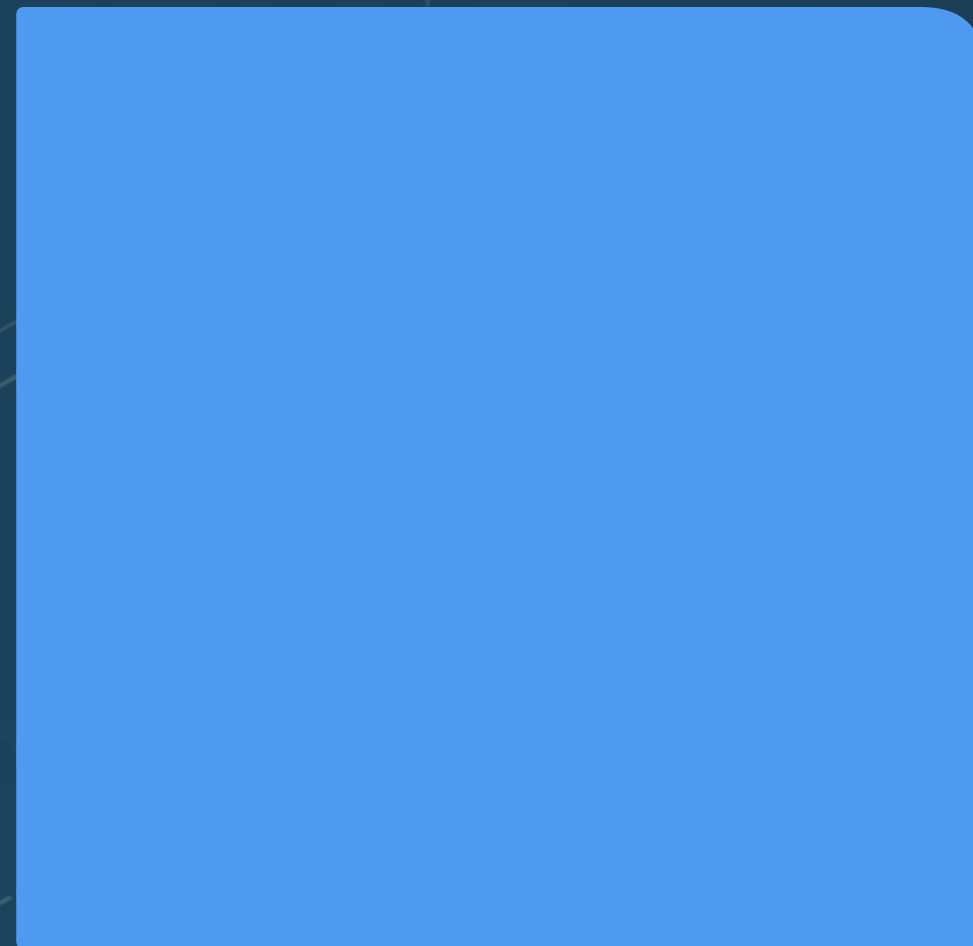
2 ✗



Employee table will have full Profile

What does a Persistence Unit represent in JPA?

18 ✓



A set of entity classes managed as a group

2 ✗



A Java class representing a database table

0 ✗



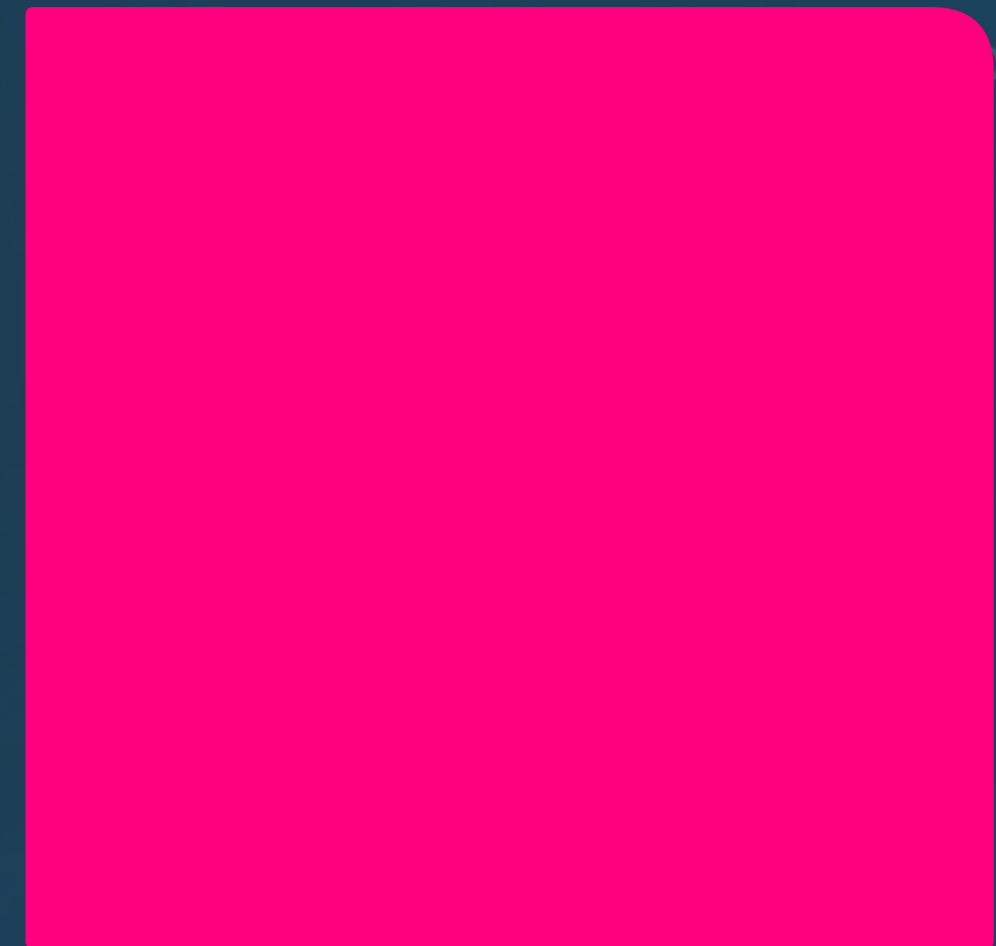
A database table representing a Java class

2

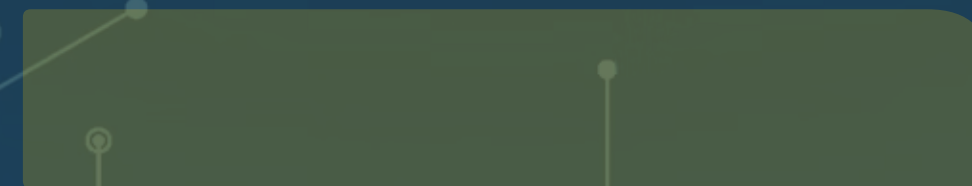


Which method is used to *delete* an entity from the database in JPA?

17 ✓



3 ✗



0 ✗

`em.drop()`

`em.delete()`

`em.remove()`

What's the purpose of *orphanRemoval* flag?

17 ✓



1 ✗

Same as Cascade Delete

2 ✗

When Child entity is deleted, then
Parent Entity would be deleted

What does FetchType.LAZY do in JPA?

17 



0 

Forces to eagerly load associated entities

0 

Automatically saves associated entities when parent entity is saved

0 

Deletes associated entities when parent entity is deleted

Quiz leaderboard



Discuss JPA practical task

Next time: Normalization, Isolation, Transaction

Questions?

